

GRAPHICS USING XLIB

(Writing programs)

Part—2

This article applies the concepts explained in Part 1 of this series with a sample program.

In case you have forgotten the concepts, a quick recap might be helpful. A client (user program) and the server communicate with each other through Xlib, which is running on either the local or a remote system. The X protocol defines each packet of information transferred in both directions, in the form of requests, replies, events and errors. I have assumed a basic knowledge of 'C' language on your part, as a requisite to learning to use Xlib.

Before we commence with programming, you should be aware of certain important conventions and concepts defined to be standard. Also remember that X is not a part of the OS, but is an application.

Windows: The window system in Xlib follows a hierarchical architecture, which makes it easy to identify the parent and child, and also control them. This hierarchy could be deep, to handle more complexities. Each window has its own coordinate system, which has the X axis (horizontal) and the Y axis (vertical) with the origin [0, 0] at the upper-left corner of the window. Coordinates are integral, in terms of pixels, and coincide with pixel centres. For a window, the origin is inside the border at the inside, upper-left corner. The X window system can support one or more screens on the monitor. A collection of screens that belongs to a user, along with the keyboard and mouse, is called a display. X provides graphics, text, and raster operations for the windows.

Function naming conventions:

1. All function names and user-visible data structures start with 'X'.
2. Macro names are entirely in upper-case letters.
3. The names of all elements in a data structure are lower-case.
4. Compound words are separated by an underscore (_).

Errors: If the function fails, it returns a zero. When Xlib detects an error, it calls an error handler, which your program can provide. If you do not provide an error handler, the error is printed, and your program terminates.

Writing programs in X

If you have read my previous article, the emphasis was on understanding how X works. The outline of an X client program could be as follows:

1. Establish a connection with the server.
2. Open a window/s.
3. Declare variables/ allocate memory.
4. Send requests and get replies from the server.
5. Handle all the errors appropriately.
6. Free all the variables.
7. Close the window/s.
8. Close the connection with the server.

Let's begin programming by following these steps, in order to avoid complexities and confusion. We will